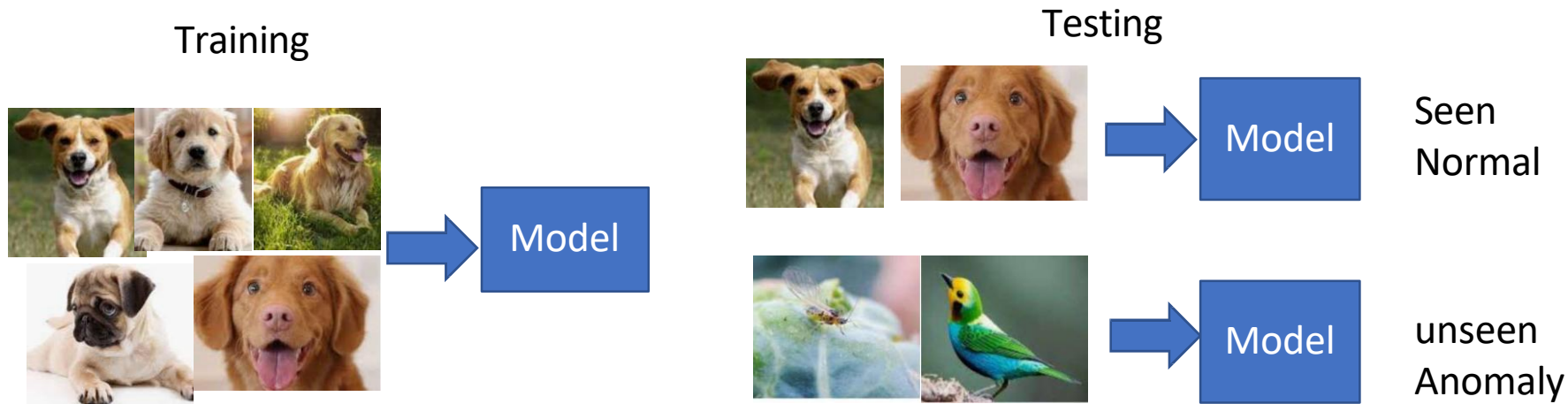


Autoencoders - Anomaly Detection

Aim

Unsupervised anomaly detection in computer vision:

Determining whether a test image belongs to the same distribution as the training images using a machine learning model.



Data

Training Set: ~140,000 human face images, each sized $64 \times 64 \times 3$ (RGB).

Testing Set:

- 10,000 normal images from the same distribution as the training set (labelled as class 0).
- 10,000 anomalous human face images from different distributions (labelled as class 1).

File provided (in zip file format):

- trainingset.npy
- testingset.npy

Autoencoder

Training & Inference Guidelines

•When to Stop Training:

Stop training once the **Mean Squared Error (MSE) loss** has converged.

•Inference Phase:

- For each test image, compute the **reconstruction error** between the input and its reconstruction.
- This reconstruction error is treated as the **anomaly score** (aka **abnormality**).

•Interpreting Anomaly Score:

A high anomaly score suggests the image likely comes from a distribution **unseen during training**. Hence, the anomaly score serves as the model's prediction of how "normal" or "anomalous" an input is.

TODO:

Complete all the key missing elements in the jupyter notebook file. It shows as **TODO** in the file

Run the pipeline to check the final reconstruction errors from the autoencoder

Change the backbone (encoder-decoder architecture) for better feature extraction and construction.